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Focus ring and apparatus for processing a substrate including the same

Abstract

A focus ring includes an inner ring on a periphery of an electrostatic chuck, and an outer ring having an inner circumferential surface surrounding an outer circumferential surface of the inner ring, wherein a lower gap between a lower end of the outer circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring is narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring, and wherein the outer ring includes a thickness-compensating portion on the inner circumferential surface of the outer ring, the thickness-compensating portion being positioned higher than an imaginary straight line between the upper end and the lower end of the inner circumferential surface of the outer ring.

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Background/Summary

CROSS-RELATED APPLICATION

(1) This application claims priority under 35 USC § 119 to Korean Patent Application No. 10-2023-0058332, filed on May 4, 2023, in the Korean Intellectual Property Office (KIPO), the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

1. Field

(2) Example embodiments relate to a focus ring and an apparatus for processing a substrate including the same. More particularly, example embodiments relate to a focus ring configured to concentrate plasma on a substrate and an apparatus for processing a substrate including the focus ring.

2. Description of the Related Art

(3) Generally, an apparatus for processing a substrate using plasma may form a layer on the substrate or etch a layer on the substrate. The apparatus may include a focus ring configured to concentrate the plasma on the substrate. The focus ring may include an inner ring and an outer ring.

SUMMARY

(4) According to example embodiments, there may be provided a focus ring. The focus ring may include an inner ring and an outer ring. The inner ring may surround a substrate on an electrostatic chuck (ESC). The outer ring may include an inner circumferential surface configured to surround an outer circumferential surface of the inner ring. A lower gap between a lower end of the outer circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring may be narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring. The outer ring may include a thickness-compensating portion on the inner circumferential surface of the outer ring. The thickness-compensating portion may be positioned higher than a straight line between the upper end and the lower end of the inner circumferential surface of the outer ring.

(5) According to example embodiments, there may be provided a focus ring. The focus ring may include an inner ring and an outer ring. The inner ring may surround a substrate on an electrostatic chuck (ESC). The outer ring may include an inner circumferential surface configured to surround an outer circumferential surface of the inner ring. A lower gap between a lower end of the outer circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring may be narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring. The inner circumferential surface of the outer ring may include a first inclined surface and a second inclined surface. The first inclined surface may be extended from the lower end of the inner circumferential surface of the outer ring at a first inclined angle with respect to a lower surface of

the outer ring. The second inclined surface may be extended from an upper end of the first inclined surface to the upper end of the inner circumferential surface of the outer ring at a second inclined angle with respect to the lower surface of the outer ring. The first inclined angle may be greater than the second inclined angle. An upper surface of the outer ring may be positioned lower than an upper surface of the inner ring.

(6) According to example embodiments, there may be provided an apparatus for processing a substrate. The apparatus may include a reaction chamber, an ESC, a plasma generator and a focus ring. The reaction chamber may receive the substrate. The ESC may be arranged in the reaction chamber to support the substrate. The plasma generator may generate plasma in the reaction chamber from a reaction gas. The focus ring may surround the substrate on the ESC. The focus ring may include an inner ring and an outer ring. The inner ring may surround the substrate. The outer ring may include an inner circumferential surface configured to surround an outer circumferential surface of the inner ring. A lower gap between a lower end of the outer circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring may be narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring. The outer ring may include a thickness-compensating portion on the inner circumferential surface of the outer ring. The thickness-compensating portion may be positioned higher than a straight line between the upper end and the lower end of the inner circumferential surface of the outer ring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features will become apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings, in which:
- (2) FIG. 1 is a cross-sectional view illustrating a capacitively coupled plasma (CCP) type apparatus for processing a substrate in accordance with example embodiments;
- (3) FIG. 2 is an enlarged cross-sectional view illustrating a focus ring of the CCP type apparatus in FIG. 1;
- (4) FIG. 3 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments;
- (5) FIG. 4 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments;
- (6) FIG. 5 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments;
- (7) FIG. 6 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments;
- (8) FIG. 7 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments;
- (9) FIG. 8 is a cross-sectional view illustrating an inductively coupled plasma (ICP) type apparatus for processing a substrate in accordance with example embodiments;
- (10) FIG. 9 is a view illustrating an electric field around a focus ring in accordance with a Comparative Example;
- (11) FIG. 10 is a view illustrating an electric field around the focus ring in FIG. 3; and
- (12) FIG. 11 is a view illustrating an electric field around the focus ring in FIG. 4.

DETAILED DESCRIPTION

(13) Hereinafter, example embodiments will be explained in detail with reference to the accompanying drawings.

(14) FIG. 1 is a cross-sectional view illustrating capacitively coupled plasma (CCP) type apparatus

for processing a substrate in accordance with example embodiments and FIG. 2 is an enlarged cross-sectional view illustrating a focus ring of the CCP type apparatus in FIG. 1.

(15) Referring to FIGS. 1 and 2, a CCP type apparatus **100** for processing a substrate may apply radiofrequency (RF) power to opposite electrodes to generate plasma from a reaction gas using an RF electric field between the electrodes. The CCP type apparatus **100** may include semiconductor fabrication apparatuses configured to process a substrate using the plasma in a reaction chamber, e.g., a deposition apparatus, an etching apparatus, an ashing apparatus, etc.

(16) The CCP type apparatus **100** may include a reaction chamber **110**, a showerhead **120**, an electrostatic chuck (ESC) **130**, a focus ring **140**, etc. That is, the CCP type apparatus **100** may include the showerhead **120** as a plasma generator.

(17) The reaction chamber **110** may have an internal space configured to receive a substrate, e.g., a semiconductor substrate S. The reaction chamber **110** may include a vacuum region configured to define a space where the plasma may be formed from a reaction gas.

(18) The showerhead **120** may be arranged at an upper region in the reaction chamber **110**. The showerhead **120** may include a plurality of injection holes configured to inject the reaction gas into the reaction chamber **110**. An RF power supply **122** may be connected to the showerhead **120**. Thus, the showerhead **120** may function as an upper electrode.

(19) The ESC **130** may be arranged at a lower region in the reaction chamber **110**, e.g., the ESC **130** and the showerhead **120** may be at opposite sides of the reaction chamber **110**. An RF power supply may be connected to the ESC **130**. Thus, the ESC **130** may function as a lower electrode. A matcher may be arranged between the RF power supply **122** and the ESC **130**. A plurality of lift holes may be vertically formed through the ESC **130**.

(20) Additionally, a plurality of lift pins may be movably inserted into the lift holes of the ESC **130**. The lift pins may support the semiconductor substrate S. The lift pins may be downwardly moved to place the semiconductor substrate S on an upper surface of the ESC **130**. The lift pins may be upwardly moved with the semiconductor substrate S on which a plasma process may be performed.

(21) The focus ring **140** may be arranged at an edge portion of the ESC **130** to surround the semiconductor substrate S. The focus ring **140** may protect an outer circumferential surface of the semiconductor substrate S from the plasma. Further, the focus ring **140** may concentrate the plasma on an upper surface of the semiconductor substrate S. When the CCP type apparatus **100** is an etching apparatus, the focus ring **140** may be partially etched by the plasma. The focus ring **140** may include an inner ring **150** and an outer ring **160**.

(22) As illustrated in FIG. 1, the inner ring **150** may be positioned on the ESC **130** and extend along its peripheral edge, e.g., the inner ring **150** may be on a periphery of the ESC **130**. The inner ring **150** may surround, e.g., a perimeter of, the semiconductor substrate S. As illustrated in FIG. 2, the inner ring **150** may have an upper surface **154**, a lower surface **156**, an inner circumferential surface, and an outer circumferential surface **152**. The inner circumferential surface of the inner ring **150** may face the semiconductor substrate S. The upper surface **154** and the lower surface **156** of the inner ring **150** may be horizontal planes opposite to each other and substantially parallel to each other, e.g., the upper and lower surfaces **154** and **156** may be parallel to a bottom of the ESC **130**. The outer circumferential surface **152** of the inner ring **150** may be a vertical plane, e.g., the outer circumferential surface **152** may be opposite the inner circumferential surface and may face away from the semiconductor substrate S. Further, a protrusion **158** may protrude from the outer circumferential surface **152** of the inner ring **150**, e.g., in a direction oriented from the inner ring **150** toward the outer ring **160**. The protrusion **158** may protrude in a radial direction of the inner ring **150**. For example, the inner ring **150** may include silicon.

(23) As illustrated in FIG. 1, the outer ring **160** may surround, e.g., a perimeter of, the inner ring **150**. As illustrated in FIG. 2, the outer ring **160** may have an upper surface **164**, a lower surface **166**, and an inner circumferential surface **162**. The upper surface **164** and the lower surface **166** of the outer ring **160** may be horizontal planes opposite to each other and substantially parallel to each

other. In example embodiments, the upper surface **164** of the outer ring **160** may be substantially coplanar with the upper surface **154** of the inner ring **150**. The lower surface **166** of the outer ring **160** may be placed on an upper surface of the protrusion **158** of the inner ring **150**. For example, the outer ring **160** may include quartz.

(24) The outer ring **160** may include an extension **163** downwardly extended from an outer side portion of the lower surface **166** of the outer ring **160**. The extension **163** of the outer ring **160** may be spaced apart from the protrusion **158** (e.g., in a radial direction) to form a space between the extension **163** and the protrusion **158**. A fixing ring **170** may be arranged in the space between the extension **163** and the protrusion **158** to fix the inner ring **150**.

(25) In example embodiments, a lower gap LG between a lower end of the outer circumferential surface **152** of the inner ring **150** and a lower end of the inner circumferential surface **162** of the outer ring **160** may be narrower than an upper gap UG between an upper end of the outer circumferential surface **152** of the inner ring **150** and an upper end of the inner circumferential surface **162** of the outer ring **160**. Thus, a sufficiently large space may be formed between the outer circumferential surface **152** of the inner ring **150** and the inner circumferential surface **162** of the outer ring **160**, e.g., the space between the outer circumferential surface **152** of the inner ring **150** and the inner circumferential surface **162** of the outer ring **160** may have an increasing width as a distance from the protrusion **158** of the inner ring **150** increases. An air current including polymers generated in a plasma process may smoothly flow through the space between the inner ring **150** and the outer ring **160**. As a result, accumulation of polymers between the inner ring **150** and the outer ring **160** may be suppressed or substantially minimized so that generation of an arc between the inner ring **150** and the outer ring **160** may also be suppressed or substantially minimized.

(26) In detail, as illustrated in FIG. 2, a lowermost end of the inner circumferential surface **162** of the outer ring **160** may contact (e.g., directly contact) a lowermost end of the outer circumferential surface **152** of the inner ring **150**. That is, a space may not be formed between the lowermost end of the inner circumferential surface **162** of the outer ring **160** and the lowermost end of the outer circumferential surface **152** of the inner ring **150**. Thus, a gap between the lowermost end of the inner circumferential surface **162** of the outer ring **160** and the lowermost end of the outer circumferential surface **152** of the inner ring **150** may be about zero. Because a space may not exist between the lowermost end of the inner circumferential surface **162** of the outer ring **160** and the lowermost end of the outer circumferential surface **152** of the inner ring **150**, polymers may not be accumulated between the lowermost end of the inner circumferential surface **162** of the outer ring **160** and the lowermost end of the outer circumferential surface **152** of the inner ring **150**.

(27) In order to smooth further the flow of the air current between the inner ring **150** and the outer ring **160**, the inner circumferential surface **162** of the outer ring **160** may be placed radially farther from the outer circumferential surface **152** of the inner ring **150** to form a large space between the inner circumferential surface **162** of the outer ring **160** and the outer circumferential surface **152** of the inner ring **150**. However, as the outer ring **160** may be, e.g., continuously, worn by plasma so that a thickness of the outer ring **160** may be gradually decreased, it may be required to periodically exchange the outer ring **160** for a new outer ring. That is, as the position of the inner circumferential surface **162** of the outer ring **160** farther from the outer circumferential surface **152** of the inner ring **150** may potentially reduce thickness of the outer ring **160**, the thin outer ring **160** may have a short lifespan so that an exchange period of the outer ring **160** may be shortened.

(28) In order to extend the lifespan of the outer ring **160** while also suppressing the arc, the outer ring **160** may include a thickness-compensating portion **161**. The thickness-compensating portion **161** may be formed on the inner circumferential surface **162** of the outer ring **160**. The thickness-compensating portion **161** may be positioned higher than an imaginary straight line L between the upper end and the lower end of the inner circumferential surface **162** of the outer ring **160** (i.e., the dashed line in FIG. 2). That is, the inner circumferential surface **162** of the outer ring **160** may correspond to an inner circumferential surface of the thickness-compensating portion **161**.

(29) In detail, the thickness-compensating portion **161** may include a first inclined surface **162a** and a second inclined surface **162b**. The first inclined surface **162a** may be upwardly extended from the lower end (e.g., lowermost end) of the inner circumferential surface **162** of the outer ring **160**. The first inclined surface **162a** may have a first inclined angle θ_1 with respect to the lower surface **166** of the outer ring **160**. The second inclined surface **162b** may be upwardly extended from an upper end of the first inclined surface **162a** to the upper end of the inner circumferential surface **162** of the outer ring **160**. The second inclined surface **162b** may have a second inclined angle θ_2 with respect to the lower surface **166** of the outer ring **160**.

(30) In example embodiments, the first inclined angle θ_1 may be greater than the second inclined angle θ_2 . For example, the first inclined angle θ_1 may be about 60° , and the second inclined angle θ_2 may be about 30° . Thus, the thickness-compensating portion **161** may have a triangular shape, e.g., so the largest surface of the triangular shape may be the imaginary straight line L. As a result, a slope of the first inclined surface **162a** may be greater than a slope of the second inclined surface **162b**.

(31) For example, a length of the first inclined surface **162a** may be shorter than a length of the second inclined surface **162b**. For example, the length of the first inclined surface **162a** may be substantially equal to or longer than the length of the second inclined surface **162b**. When the length of the first inclined surface **162a** is shorter than the length of the second inclined surface **162b**, a connection point between the first inclined surface **162a** and the second inclined surface **162b**, i.e., the upper end of the first inclined surface **162a** or the lower end of the second inclined surface **162b**, may be positioned closer to the lower surface **166** of the outer ring **160** than to the upper surface **164** of the outer ring **160**.

(32) FIG. 3 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments.

(33) A CCP type apparatus **100a** of example embodiments may include elements substantially the same as those of the CCP type apparatus **100** in FIG. 2, except for an outer ring of a focus ring. Thus, same reference numerals refer to the same elements and any further description with respect to the same elements is omitted herein for brevity.

(34) Referring to FIG. 3, a lowermost end of the inner circumferential surface **162** of an outer ring **160a** may be spaced apart from the lowermost end of the outer circumferential surface **152** of the inner ring **150** (e.g., in the radial direction to expose a portion of the upper surface of the protrusion **158**). Thus, the lower gap LG may be formed between the lowermost end of the inner circumferential surface **162** of the outer ring **160a** and the lowermost end of the outer circumferential surface **152** of the inner ring **150**. The lower gap LG may be about 0.04 mm to about 2 mm.

(35) FIG. 4 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments.

(36) A CCP type apparatus **100b** of example embodiments may include elements substantially the same as those of the CCP type apparatus **100** in FIG. 2, except for an outer ring of a focus ring. Thus, same reference numerals refer to the same elements and any further description with respect to the same elements is omitted herein for brevity.

(37) Referring to FIG. 4, an upper surface **164b** of an outer ring **160b** may be positioned lower than the upper surface **154** of the inner ring **150**. That is, a height difference V may be formed between the upper surface **164b** of the outer ring **160b** and the upper surface **154** of the inner ring **150**. For example, the height difference V may be at least about 1 mm.

(38) Because the upper surface **164b** of the outer ring **160b** may be positioned lower than the upper surface **154** of the inner ring **150**, a space through which the air current may smoothly flow may be formed over the outer ring **160b**. Thus, the accumulation of the polymers between the inner ring **150** and the outer ring **160b** may be suppressed or substantially minimized so that generation of the arc between the inner ring **150** and the outer ring **160b** may also be suppressed.

(39) FIG. 5 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments.

(40) A CCP type apparatus **100c** of example embodiments may include elements substantially the same as those of the CCP type apparatus **100** in FIG. 2, except for an outer ring of a focus ring. Thus, same reference numerals refer to the same elements and any further description with respect to the same elements is omitted herein for brevity.

(41) Referring to FIG. 5, the thickness-compensating portion **161** of example embodiments may further include a third inclined surface **162c**. The third inclined surface **162c** may be extended from the upper end of the first inclined surface **162a** to the lower end of the second inclined surface **162b**. The third inclined surface **162c** may have a third inclined angle θ_3 with respect to the lower surface **166** of the outer ring **160**. The third inclined angle θ_3 may be less than the first inclined angle θ_1 and greater than the second inclined angle θ_2 . Alternatively, the outer ring **160** may include at least four inclined surfaces and/or the structure in FIG. 3 and/or the structure in FIG. 4 with at least three inclined surfaces defining the inclined surface **162**.

(42) FIG. 6 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments.

(43) A CCP type apparatus **100d** of example embodiments may include elements substantially the same as those of the CCP type apparatus **100** in FIG. 2, except for an outer ring of a focus ring. Thus, same reference numerals refer to the same elements and any further description with respect to the same elements is omitted herein for brevity.

(44) Referring to FIG. 6, a thickness-compensating portion **161d** of example embodiments may include a curved inner circumferential surface **162d** connected between the upper end and the lower end of the inner circumferential surface **162** of the outer ring **160d**. That is, the thickness-compensating portion **161d** may have an arc shape. Alternatively, the focus ring of example embodiments may have the structure in FIG. 3 and/or the structure in FIG. 4 with the curved inner circumferential surface **162d**.

(45) FIG. 7 is an enlarged cross-sectional view illustrating a focus ring of a CCP type apparatus in accordance with example embodiments.

(46) A CCP type apparatus **100e** of example embodiments may include elements substantially the same as those of the CCP type apparatus **100** in FIG. 2, except for an outer ring of a focus ring. Thus, same reference numerals refer to the same elements and any further description with respect to the same elements is omitted herein for brevity.

(47) Referring to FIG. 7, a thickness-compensating portion **161e** of example embodiments may include a first inclined surface **162-1** and a second inclined surface **162-2**. The first inclined surface **162-1** may be upwardly extended from the lower end of the inner circumferential surface **162e** of the outer ring **160e**. The first inclined surface **162-1** may have a fourth inclined angle θ_4 with respect to the lower surface **166** of the outer ring **160e**. The second inclined surface **162-2** may be upwardly extended from the upper end of the first inclined surface **162-1** to the upper end of the inner circumferential surface **162e** of the outer ring **160e**. The second inclined surface **162-2** may have a fifth inclined angle θ_5 with respect to the lower surface **166** of the outer ring **160e**.

(48) Further, a length of the first inclined surface **162-1** may be longer than a length of the second inclined surface **162-2**. For example, the length of the first inclined surface **162-1** may be substantially equal to or less than the length of the second inclined surface **162-2**. When the length of the first inclined surface **162-1** is longer than the length of the second inclined surface **162-2**, a connection point between the first inclined surface **162-1** and the second inclined surface **162-2**, i.e., the upper end of the first inclined surface **162-1** or the lower end of the second inclined surface **162-2**, may be positioned closer to the upper surface **164** of the outer ring **160e** than the lower surface **166** of the outer ring **160e**. Thus, an upper gap UGe between the upper end of the outer circumferential surface **152** of the inner ring **150** and the upper end of the inner circumferential surface **162e** of the outer ring **160e** may be narrower than the upper gap UG between the upper end

of the outer circumferential surface **152** of the inner ring **150** and the upper end of the inner circumferential surface **162** of the outer ring **160** in FIG. 2. As a result, a volume of a space between the inner ring **150** and the outer ring **160e** in FIG. 7 may be smaller than a volume of the space between the inner ring **150** and the outer ring **160** in FIG. 2.

(49) FIG. 8 is a cross-sectional view illustrating an inductively coupled plasma (ICP) type apparatus for processing a substrate in accordance with example embodiments.

(50) Referring to FIG. 8, an ICP type apparatus **200** may generate plasma from a reaction gas using a magnetic field induced by a coil antenna. An RF power applied to the coil antenna may be transmitted to the reaction gas through a dielectric window. The ICP type apparatus **200** may include semiconductor fabrication apparatuses configured to process the substrate using plasma in a reaction chamber, e.g., a deposition apparatus, an etching apparatus, an ashing apparatus, etc.

(51) The ICP type apparatus **200** may include a reaction chamber **210**, an antenna **224**, a dielectric window **220**, an ESC **230**, a focus ring **140**, etc. That is, the ICP type apparatus **200** may include the antenna **224** (e.g., a coil antenna) and the dielectric window **220** as the plasma generator.

(52) The reaction chamber **210** may have an internal space configured to receive the substrate, e.g., the semiconductor substrate S. The reaction chamber **210** may have a vacuum region for defining a space where the plasma may be formed from the reaction gas.

(53) The antenna **224** may be arranged on the upper surface of the reaction chamber **210**. An RF power supply **222** may be connected to the antenna **224**. A magnetic field induced by the antenna **224** may be applied to the reaction gas injected into the reaction chamber **210** to generate the plasma.

(54) The dielectric window **220** may be arranged under the antenna **224**. The dielectric window **220** may include a dielectric material. The dielectric window **220** may transmit the RF power in the antenna **224** into the reaction chamber **210**. Further, the dielectric window **220** may inject the reaction gas into the reaction chamber **210**.

(55) The ESC **230** may be arranged at a lower region in the reaction chamber **210**. An RF power supply may be connected to the ESC **230**. A matcher may be arranged between the RF power supply and the ESC **230**.

(56) A plurality of lift holes may be vertically formed through the ESC **230**. Additionally, a plurality of lift pins may be movably inserted into the lift holes of the ESC **20**. The lift pins may support the semiconductor substrate S. The lift pins may be downwardly moved to place the semiconductor substrate S on an upper surface of the ESC **230**. The lift pins may be upwardly moved with the semiconductor substrate S on which a plasma process may be performed.

(57) The focus ring **140** may have a structure substantially the same as the structure of the focus ring **140** in FIG. 2. Thus, any further descriptions with respect to the focus ring **140** are omitted herein for brevity. Alternatively, the focus ring **140** may have the structure in FIG. 3, the structure in FIG. 4, the structure in FIG. 5, the structure in FIG. 6, or the structure in FIG. 7.

(58) FIG. 9 is a view illustrating an electric field around a focus ring in accordance with a Comparative Example, FIG. 10 is a view illustrating an electric field around the focus ring in FIG. 3, and FIG. 11 is a view illustrating an electric field around the focus ring in FIG. 4.

(59) The focus ring in FIG. 9 may have a uniform gap between an inner ring and an outer ring. In contrast, the focus rings in FIGS. 10 and 11 may include the thickness-compensating portion, according to example embodiments.

(60) Electric fields applied to the focus rings in FIGS. 9 to 11 were measured. Amounts of polymers deposited on the focus rings were measured. Results are shown in following Table.

(61) TABLE-US-00001 Comparative Focus ring Focus ring Example in FIG. 3 in FIG. 4 Ratio of polymers on focus rings 8.59% 2.03% 2.21% with respect to total amount of polymers (%) Maximum electric field (E-field 242.7 88.8 88.0 (kV/m)

(62) As shown in the Table above, it can be noted that a deposition ratio of the polymer on the focus ring was decreased from 8.59% to 2.03% or 2.21%. Thus, the non-uniform (i.e., non-constant

width) space formed between the inner ring and the outer ring, according to example embodiments, reduces the deposition ratio of the polymer on the focus ring.

(63) Further, it can be noted that the electric field applied to the focus rings may be decreased from 242.7 to 88.8 or 88.0. Thus, the space formed between the inner ring and the outer ring, according to example embodiments, reduces a concentration of the electric field by the focus ring.

(64) By way of summation and review, polymers generated in a plasma process may be accumulated in a space between the inner ring and the outer ring of a focus ring, such that the accumulated polymers may cause an arcing.

(65) In contrast, example embodiments provide a focus ring capable of suppressing generation of an arcing. Example embodiments also provide an apparatus for processing a substrate including the above-mentioned focus ring.

(66) That is, according to example embodiments, the lower gap between the lower end of the outer circumferential surface of the inner ring and the lower end of the inner circumferential surface of the outer ring may be narrower than the upper gap between the upper end of the outer circumferential surface of the inner ring and the upper end of the inner circumferential surface of the outer ring so that a sufficient space may be formed between the inner ring and the outer ring. Thus, an air current may smoothly flow through the space between the inner ring and the outer ring. As a result, an accumulation of polymers in the air current between the inner ring and the outer ring may be suppressed so that an arc between the inner ring and the outer ring may also be suppressed.

(67) Further, a thickness-compensating portion on the inner circumferential surface of the outer ring may be positioned higher than the straight line between the upper end and the lower end of the inner circumferential surface of the outer ring so that the thickness-compensating portion may provide the outer ring with a sufficiently thick thickness. Thus, a lifespan of the outer ring worn by the plasma may be extended.

(68) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A focus ring, comprising: an inner ring on a periphery of an electrostatic chuck; and an outer ring having an inner circumferential surface surrounding an outer circumferential surface of the inner ring, wherein a lower gap between a lower end of the outer circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring is narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring, and wherein the outer ring includes a thickness-compensating portion on the inner circumferential surface of the outer ring, the thickness-compensating portion being positioned higher than an imaginary straight line between the upper end and the lower end of the inner circumferential surface of the outer ring.

2. The focus ring as claimed in claim 1, wherein the thickness-compensating portion includes: a first inclined surface extended from the lower end of the inner circumferential surface of the outer ring at a first inclined angle with respect to a lower surface of the outer ring; and a second inclined surface extended from an upper end of the first inclined surface to the upper end of the inner

circumferential surface of the outer ring at a second inclined angle with respect to the lower surface of the outer ring.

3. The focus ring as claimed in claim 2, wherein the first inclined angle is greater than the second inclined angle.

4. The focus ring as claimed in claim 2, wherein the thickness-compensating portion further includes a third inclined surface extended from the upper end of the first inclined surface to a lower end of the second inclined surface at a third inclined angle with respect to the lower surface of the outer ring.

5. The focus ring as claimed in claim 4, wherein the third inclined angle is less than the first inclined angle and greater than the second inclined angle.

6. The focus ring as claimed in claim 1, wherein the thickness-compensating portion includes a curved inner circumferential surface connected between the upper end and the lower end of the inner circumferential surface of the outer ring.

7. The focus ring as claimed in claim 1, wherein the lower end of the inner circumferential surface of the outer ring is in contact with the lower end of the outer circumferential surface of the inner ring.

8. The focus ring as claimed in claim 1, wherein the lower end of the inner circumferential surface of the outer ring is spaced apart from the lower end of the outer circumferential surface of the inner ring.

9. The focus ring as claimed in claim 1, wherein an upper surface of the outer ring is substantially coplanar with an upper surface of the inner ring.

10. The focus ring as claimed in claim 1, wherein an upper surface of the outer ring is positioned lower than an upper surface of the inner ring.

11. A focus ring, comprising: an inner ring on a periphery of an electrostatic chuck; and an outer ring having an inner circumferential surface surrounding an outer circumferential surface of the inner ring, wherein a lower gap between a lower end of the outer circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring is narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring, wherein the inner circumferential surface of the outer ring includes: a first inclined surface extended from the lower end of the inner circumferential surface of the outer ring at a first inclined angle with respect to a lower surface of the outer ring, and a second inclined surface extended from an upper end of the first inclined surface to the upper end of the inner circumferential surface of the outer ring at a second inclined angle with respect to the lower surface of the outer ring, wherein the first inclined angle is greater than the second inclined angle, and wherein an upper surface of the outer ring is positioned lower than an upper surface of the inner ring.

12. The focus ring as claimed in claim 11, wherein the lower end of the inner circumferential surface of the outer ring is in contact with the lower end of the outer circumferential surface of the inner ring.

13. The focus ring as claimed in claim 11, wherein the lower end of the inner circumferential surface of the outer ring is spaced apart from the lower end of the outer circumferential surface of the inner ring.

14. An apparatus for processing a substrate, the apparatus comprising: a reaction chamber configured to receive the substrate; an electrostatic chuck in the reaction chamber, the electrostatic chuck being configured to support the substrate; a plasma generator configured to generate plasma from a reaction gas in the reaction chamber; and a focus ring on a periphery of the electrostatic chuck, the focus ring being configured to surround the substrate, and the focus ring including: an inner ring on the periphery of the electrostatic chuck and configured to surround the substrate, and an outer ring having an inner circumferential surface configured to surround an outer circumferential surface of the inner ring, wherein a lower gap between a lower end of the outer

circumferential surface of the inner ring and a lower end of the inner circumferential surface of the outer ring is narrower than an upper gap between an upper end of the outer circumferential surface of the inner ring and an upper end of the inner circumferential surface of the outer ring, and wherein the outer ring includes a thickness-compensating portion on the inner circumferential surface of the outer ring, the thickness-compensating portion being positioned higher than an imaginary straight line between the upper end and the lower end of the inner circumferential surface of the outer ring.

15. The apparatus as claimed in claim 14, wherein the plasma generator includes a showerhead configured to inject the reaction gas into the reaction chamber.

16. The apparatus as claimed in claim 14, wherein the plasma generator includes: an antenna arranged on an upper surface of the reaction chamber; and a dielectric window arranged under the antenna.

17. The apparatus as claimed in claim 14, wherein the thickness-compensating portion includes: a first inclined surface extended from the lower end of the inner circumferential surface of the outer ring at a first inclined angle with respect to a lower surface of the outer ring; and a second inclined surface extended from an upper end of the first inclined surface to the upper end of the inner circumferential surface of the outer ring at a second inclined angle with respect to the lower surface of the outer ring, the first inclined angle being greater than the second inclined angle.

18. The apparatus as claimed in claim 14, wherein the lower end of the inner circumferential surface of the outer ring makes is in contact with the lower end of the outer circumferential surface of the inner ring.

19. The apparatus as claimed in claim 14, wherein an upper surface of the outer ring is substantially coplanar with an upper surface of the inner ring.

20. The apparatus as claimed in claim 14, wherein an upper surface of the outer ring is positioned lower than an upper surface of the inner ring.
